METRIC
MIL-M-24736/2(NAVY)
31 August 1989

MILITARY SPECIFICATION SHEET

MULTIPLEXERS, DEMULTIPLEXERS, MULTIPLEXERS/DEMULTIPLEXERS (MULDEMS), DEMULTIPLEXER, TIME-DIVISION, FIBER OPTIC INTERFACEABLE, SHIPBOARD, 1:8 CHANNELS, 311 MBPS PER OUTPUT CHANNEL (METRIC)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-24736.

CLASSIFICATION

Equipment function: D (Time-division demultiplexer).
Signal type: A (Electrical).
Channel configuration: A (1 to 16 channels).
Analog subchannel class: J (Not applicable).
Digital subchannel class: H (Not greater than 1 gigabit per second).
Style: A (Microcircuit: integrated circuit (IC) or hybrid, package and mounting technique).

AMSC N/A FSC 6008
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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RELIABILITY: MTBF greater than 40,000 hours.

DESIGN AND CONSTRUCTION

Case: 52-pin leadless chip carrier, quad flat pack.
4.9 cm X 4.9 cm X 1.5 cm. See figure 1 for pin diagram.

GENERAL PERFORMANCE CHARACTERISTICS

Demultiplexers: 1 input channel, 8 output channels.

Analog subchannel output interface element: N/A.

Digital subchannel output interface elements: maximum signal rate of 311.04 megabits per second (Mbps) per output channel.

Time-division demultiplexer/control element: The input information shall be organized in accordance with the frame structure shown on figure 2.

Demultiplexer input interface element: The composite input data rate shall be framed with the appropriate control information at a maximum line rate of 2488.320 Mbps.

DETAILED PERFORMANCE CHARACTERISTICS

Prime power (dc): minus 5.2 volts, minus 2.0 volts.

Dc prime power variation:
plus or minus 0.3 volt for 5.2-volt supply;
plus or minus 0.1 volt for 2.0-volt supply.

Current load (dc): 0.4 to 0.7 amp for minus 5.2-volt supply;
0.1 to 0.3 amp for minus 2.0-volt supply.

Frame synchronization: (See figure 3.) The frame recovery circuits look for 3 F1's followed by 3 F2's. The byte clock out (BYCKO) and parallel byte data out (DO_1-DO_8) become invalid on the falling edge of OOFN and become valid when F1 changes to F2. The frame recovery circuits align the received serial data on byte boundaries for demultiplexing by controlling the timing generator. The byte boundary alignment is based on the specific F1 and F2 byte recognition.

The falling edge of OOFN must occur at least 4 byte clock periods before F1 changes to F2. The pulse width of OOFN shall be not less than 1 byte clock period.

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The frame detector monitors every demultiplexed output byte. If 3 F1 bytes followed by 3 F2 bytes are detected, then a frame confirmation signal is sent to the output FP. The rising edge of the FP pulse occurs 2 byte clock periods after F1 changes to F2 on the demultiplexer parallel data outputs. The FP pulse width is one byte clock period (see figure 3).

The frame detection circuitry also disables the frame recovery circuits once 3 F1 bytes are followed by 3 F2 bytes. The frame detector sends an FP pulse every frame when 3 F1's are followed by 3 F2's independent of the condition of the input OOFN.

Frame loss and reacquisition time: N/A.

Alarm functions: N/A.

Framing: See figure 3.

Error control: N/A.

Interchannel crosstalk: N/A.

Optical performance: N/A.

Channel performance:

Analog subchannels: N/A.
Digital channels:
Input signal voltage, high: minus 1.0 to minus 0.6 volt;
Input signal voltage, low: minus 2.0 to minus 1.6 volts.

Interface characteristics:

Analog subchannels: N/A.
Digital channels: Signal format shall be in accordance with figure 2. The rise and fall times and the clock and data relationship shall be in accordance with figure 3. The data rate shall be a maximum of 311.04 Mbps per output channel, and a maximum input line rate of 2488.320 Mbps.

Multiplexed signal interface:

TDM signal output: N/A.
TDDM signal input:
Electrical interface:
Output signal voltage, high: minus 0.9 to minus 0.6 volt.
Output signal voltage, low: minus 2.0 to minus 1.6 volts.
Optical interface: N/A.

Timing signals: 2.5 gigahertz (GHz) clock. The clock input signal level shall be from 0.6 to 1.2 volts. The clock output signal level shall be 0.4 volt (peak to peak), centered at minus 1.5 volts.

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Maintenance orderwire: N/A.
Other interfaces: N/A.

ENVIRONMENTAL REQUIREMENTS

Temperature range: 1 (operating - 0 to plus 85°C;
storage - minus 62 to plus 85°C).
Nuclear radiation resistance: N/A.

MECHANICAL PROPERTIES:

Connector pull-out force: N/A.
Cable pull-out force: N/A.
Cable twist: N/A.
Cable seal flexing: N/A.
Fiber dynamic strength: N/A.

ELECTROMAGNETIC INTERFERENCE SUPPRESSION: N/A.

SAFETY: N/A.

SECURITY: N/A.

ENVIRONMENTAL TESTS

Explosive atmosphere: N/A.

Electromagnetic interference suppresion: N/A.

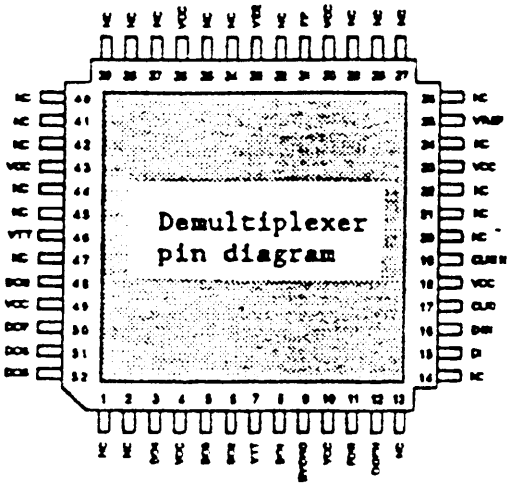
PART OR IDENTIFYING NUMBER: D24736/02-001.

PART DESIGNATOR: DAAJHA.

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Navy - SH
(Project 6008-N001-02)

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Name	Function
DO_1 - DO_8	1:8 Demux parallel ECL outputs
CLKI, CLKIN	High speed differential clock inputs
BYCKO	1:8 Demux divide by 8 clock (ECL output)
DI, DIN	High speed 1:8 Demux serial data input
DOFN	Frame Recovery Enable (ECL input)
FDIS	Frame Recovery Disable (ECL input)
FP	Frame Detection Confirmation (ECL output)
NC	No Connect

NOTE:
1. Tabs indicate position of leadless contact only and are not real structures.

FIGURE 1. Demultiplexer pin diagram.

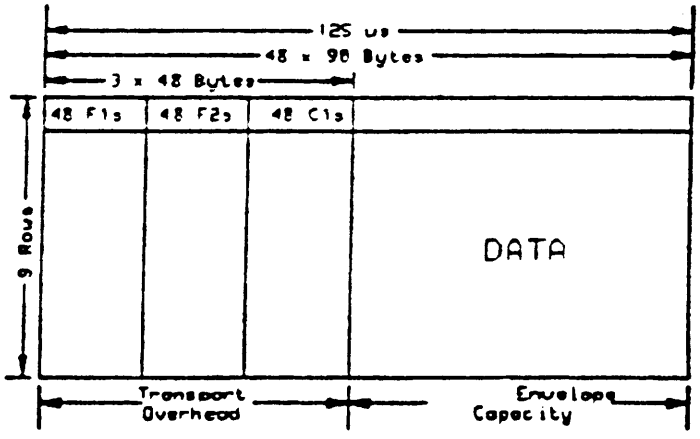
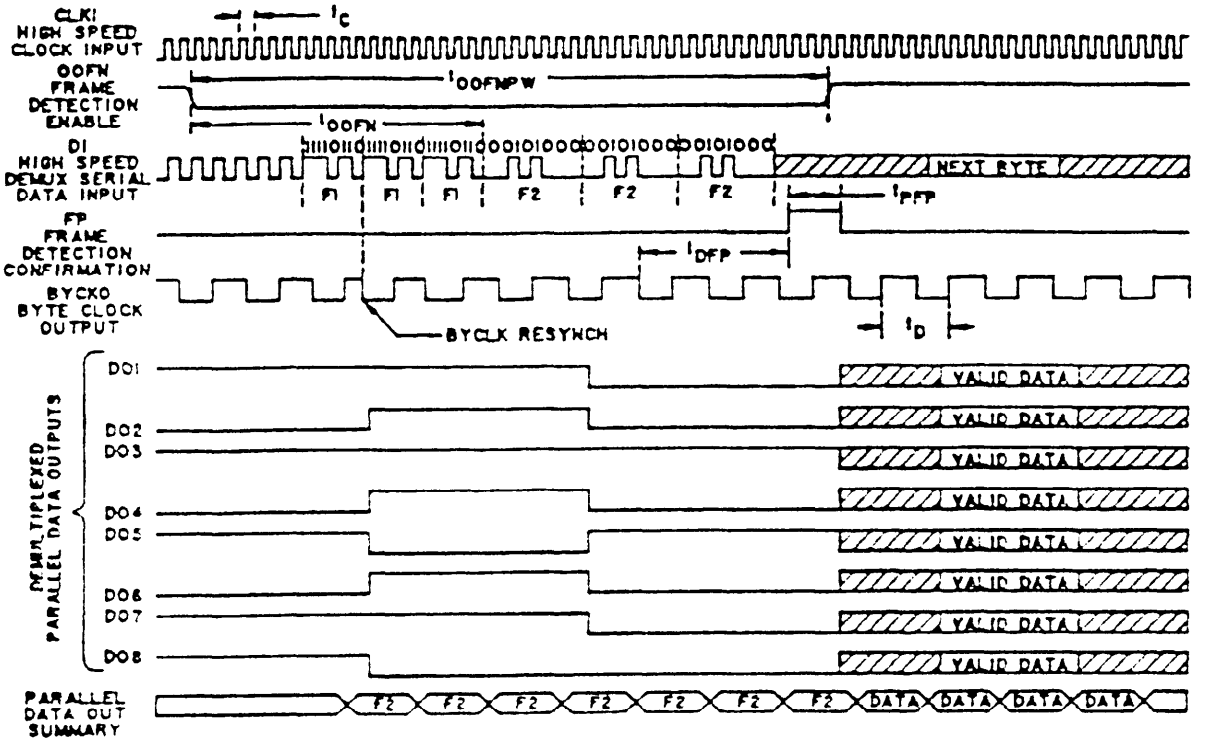


FIGURE 2. Frame structure.

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NOTE:

1. The parallel data outputs only begin showing valid data after the last F2 of the framing sequence. The example waveforms shown above use 3 F1's and 3 F2's framing sequence for convenience. Thus, in this example, valid data is output after the third F2 in the framing sequence.

FIGURE 3. Clock and data relationship.

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PARAMETER	DESCRIPTION	GUARANTEED WORST CASE (nanoseconds)
t_C 1/	CLK period	0.4
t_D	Byte clock period ($t_D = t_C \times 8$)	3.2
t_{DFP}	FP rising edge from parallel data output change from F1 to F2 ($t_{DFP} = t_D \times 2$)	6.4
t_{PFP}	FP pulse width ($t_{PFP} = t_D$)	3.2
t_{OOFN}	OOFN falling edge before F1 changes to F2 ($t_{OOFN} = t_D \times 4$)	12.8
t_{OOFNPW}	OOFN pulse width ($t_{OOFNPW} = t_D$)	3.2

1/ If t_C changes, all the remaining parameters change as indicated by the equations.

FIGURE 3. Clock and data relationship - Continued.

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